HIST.

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Q) Why was Gupta period classified as 'classical age' of India?

Ans. • Gupta rulers were devotees of Vishnu which is often reflected in their coins but they also supported other traditions and patronized Buddhist institutions including Nalanda University and several ~~Buddhist~~ Viharas.

• It was also the time when knowledge from previous eras was consolidated into numerous text. such as Kalidasa's work in sanskrit and many major Puranas.

• Aryabhata and Varahmihira recorded major advancements in mathematics and Astronomy. Many medical texts were compiled during their time.

- Metallurgy also progressed as we can see the rust resistant Iron Pillar at Mehrauli

- Chandragupta II also patronized many scholar artists and scientists lading to cultural and intellectual ~~age~~ growth.

Q) Write the achievements of the following under Guptas:

(i) Aryabhata

- He authored a short treaties of Maths and Astronomy called Aryabhatiya.

- He gave formulas to calculate the motions of the sun, the moon and the Planets and Proposed that the Earth spins on its axis which explains the alteration of day and night.

- He gave the length of the year as 365 days, 6 hrs, 12 mins and 30 secs.

- He also provided a fair estimate of the size of the Earth and correct explaination of solar and lunar eclipses.

- In maths, he described a number of techniques of calculations and equation solving.

(ii) Varahmihira

- Varahmihira's encyclopedia 'Brihat Samhita', covered a wide range of subjects from astronomy and astrology for eg. Weather Forecasting, Town Planning and even farming

- His ability to observe the world, apply logical reasoning and combine it with traditional knowledge made him Pioneer in Science.

(iii) Kalidasa

- He is famous for his contributions in Sanskrit literature and refined poetry.

· One of his celebrated composition is 'meghadootam'.

GEO.

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Q) How has the Monsoon climate influence unity in India?

OR

How has climate influenced life in India?

- It defines the the season in India that is Winter, Summer, advancing monsoon and retreating monsoon.

- Indian agriculture is mostly rain fed and is therefore dependent on monsoon.

- The major vegetation is tropical deciduous forest which is influenced by the monsoon.

- Indian food and dresses also reflects the climate of India.

- Indian literature, dance, music are all related to seasons or climate.

Q) What is Disaster?

- They are natural or manmade causes destruction to life and property.
Eg - cyclone, wars, floods, landslide, etc.

Climatic disaster

- They are certain disaster caused due to sudden climatic changes.
Eg - floods, drought, cyclones.

Floods / Deluge

- It occurs when there is a sudden rise in the level of water or a water body particullarly a river. It

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submerges low lying areas causing wide spread destructions. It damages life and property and spoils agricultural land. It also disrupts transportation and communication links.

MATH

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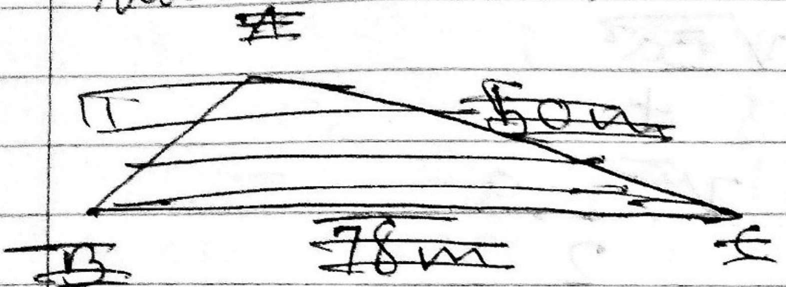
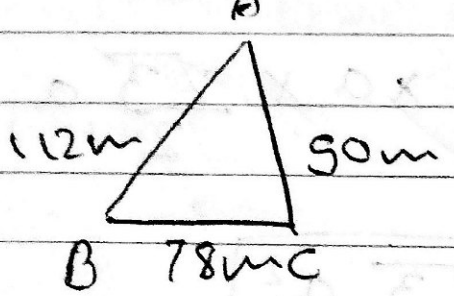
Q) Find the area of a Δ field whose sides are 78m, 50m, & 112m.





Let $a = 78m$; $b = 50m$; $c = 112m$

$$s = \frac{1}{2}(a+b+c)$$

$$= \frac{1}{2}(78+50+112)\,m$$

$$= \frac{1}{\cancel{2}} \times \cancel{240}^{120}\,m = 120m$$

Area of $\Delta ABC = \sqrt{s(s-a)(s-b)(s-c)}$

$$= \sqrt{120(120-78)(120-50)(120-112)}$$

$$= \sqrt{120 \times 42 \times 70 \times 8}\ m^2$$

$$= \sqrt{10 \times 4 \times 3 \times 7 \times 3 \times 2 \times 7 \times 10 \times 4 \times 2}$$

$$= 10 \times 4 \times 3 \times 7 \times 2\ m^2$$

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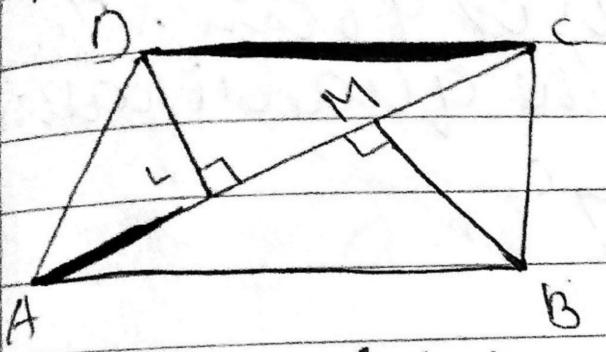
$= \boxed{1680\ m^2}$

Area of a Quadrilateral:



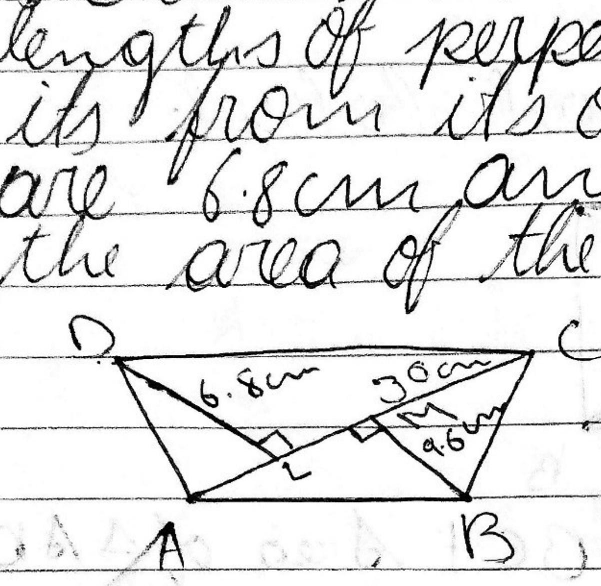
$= \text{area of } \triangle ABC + \text{Area of } \triangle ADC$

$= \left(\frac{1}{2} \times AC \times BM\right) + \left(\frac{1}{2} \times AC \times DL\right)$

$= \frac{1}{2} \times AC \times (BM + DL)$

$= \frac{1}{2} \times \text{diagonal} \times (\text{sum of its offsets})$

Q) A diagonal of a Quadrilateral is 30cm in length and the lengths of perpendicular to its from its opposite vertices are 6.8cm and 9.6cm. Find the area of the quadrilateral.



Area of a Quadrilateral ABCD

$= \frac{1}{2} \times \text{diagonal} (\text{Sum of its offsets})$

$= \frac{1}{2} \times 30 \times (6.8 + 9.6) \text{ cm}^2$

$= \frac{1}{\cancel{2}_1} \times \overset{15}{\cancel{30}} \times 16.4 \text{ cm}^2$

$= 15 \times 16.4 \text{ cm}^2$

$= \overset{3}{\cancel{15}} \times \frac{\overset{82}{\cancel{164}}}{\cancel{10}_{\cancel{2}_1}} \text{ cm}^2 = \boxed{246 \text{ cm}^2}$

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EX-20D

1) Find the area of the triangle in which

(iii) $b = 8$ dm & $h = 35$ cm.

Area of a $\Delta = \left(\frac{1}{2} \times b \times h\right)$

$= \left(\frac{1}{\cancel{2}_1} \times \overset{40}{\cancel{80}} \times 35\right) cm^2$

$= \boxed{140\ cm^2}$

2) Find the ht. of a Δ having an area of $72\ cm^2$ and base 16 cm.

Area of a $\Delta = \frac{1}{2} \times b \times h$

$\Rightarrow 72 = \frac{1}{2} \times 16 \times h$

$\Rightarrow 72 \times 2 = 16 \times h$

$\Rightarrow \frac{\overset{9}{\cancel{72}} \times \cancel{2}^1}{\underset{8_1}{\cancel{16}}} = h$

$\therefore h = \boxed{9 cm}$

6) The area of a triangular region is 129.5 cm^2. If one of the sides containing the right angle is 14.8 cm, find the other one.

Let the length of the other leg be h cm.

Then, area of the $\Delta = \left(\frac{1}{2} \times \frac{\overset{74}{\cancel{148}}}{10} \times h\right) \text{cm}$

$= 7.4h \text{ cm}^2$

But area of $\Delta = 129.5 \text{ cm}^2$

$\therefore 7.4h = 129.5$

$\Rightarrow h = \frac{129.5}{7.4}$

$\Rightarrow h = \frac{\overset{17.5}{\cancel{1295}} \times \cancel{10}}{\underset{1}{\cancel{74}} \times \cancel{10}} = 17.5 \text{ cm}$

CHEM.

Metal and Non Metal

Element

Metal | Nonmetal | Metaloid

Metal- Metals release electron easily

Def- The elements which releases electron easily are metals.

Nonmetals- The elements which accepts the electron easily.

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Physical Properties of Metals

1. Malability - The Properties due to which a metal can be beaten into thin sheet is called malability. Eg- Aluminium (foil)

2. Ductility - The property due to which a metal can be drawn into wire is called ductility.

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(3) Conductivity - The metals can conduct electricity as well as transfer the heat is called electrical conductivity and thermal conductivity. So metals used to make cookwars.

(4) Hardness - Most of the metals are hard and Solid due to metallic bond.
Eg - Iron
Exceptional - Na, K, Li, Hg etc.

(5) Tensile strength - Metal has high tensile strength so metals are used to make strong structures.
Iron is used in buildings, Machines, ships etc.

Non metals are not having Malleablity, ductility, conductivity, Hardness, Tensility strength.

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Lusture- Metals are lustrous eg gold, silver, iron, aluminium, etc. Due to this property silver and golds are used in construction of ornaments

Sonority - The property by which metals can produce a ringing sound. Therefore its used in bells.